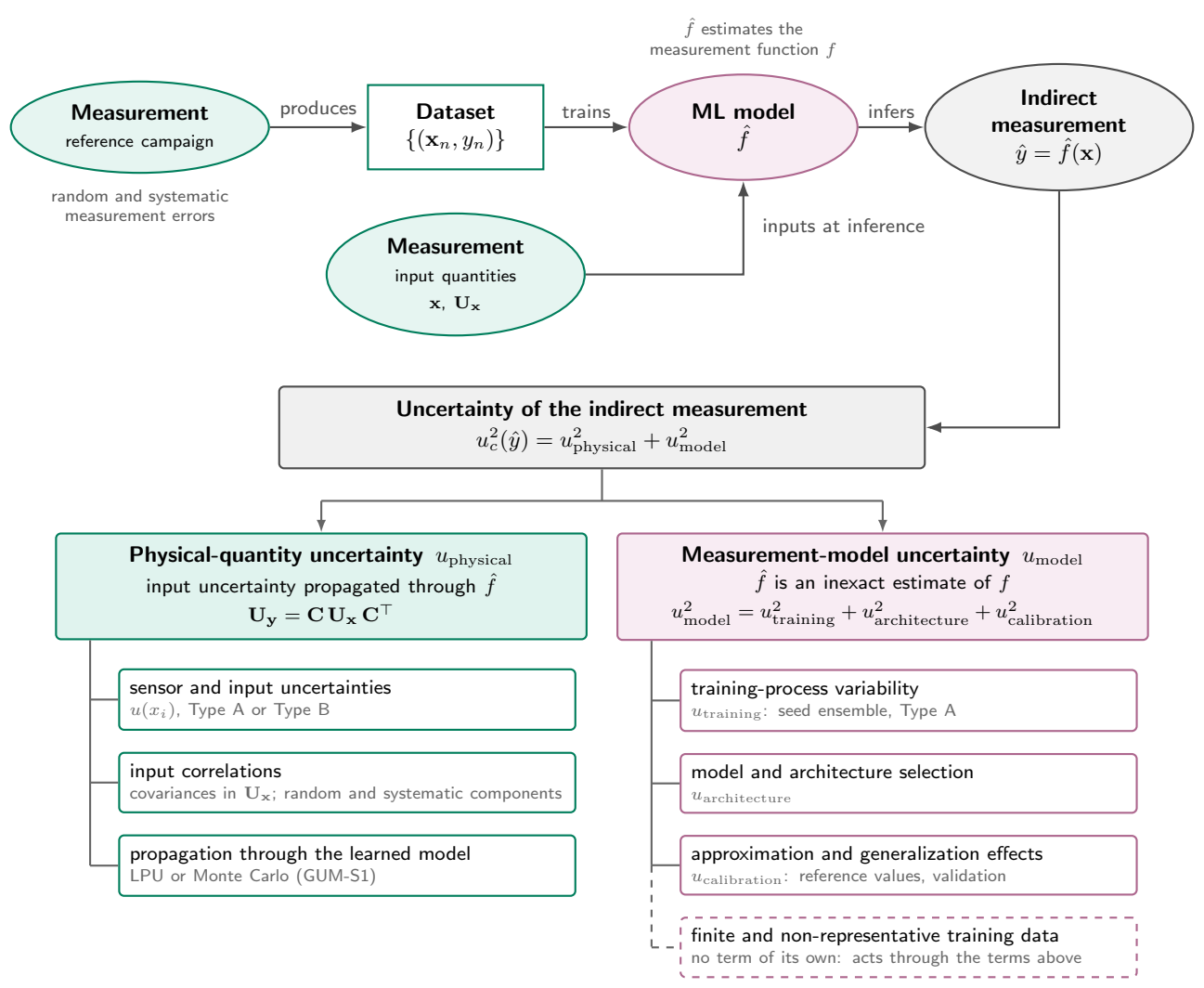

$\hat{f}$ estimates the measurement function f
Measurement
reference campaign
random and systematic measurement errors
produces
Dataset
$\{(\mathbf{x}_n, y_n)\}$
trains
ML model
$\hat{f}$
infers
Indirect measurement
$\hat{y} = \hat{f}(\mathbf{x})$
Measurement
input quantities
$\mathbf{x}$, $\mathbf{U_x}$
inputs at inference
Uncertainty of the indirect measurement
$u_c^2(\hat{y}) = u_{\text{physical}}^2 + u_{\text{model}}^2$
Physical-quantity uncertainty u_{physical}
input uncertainty propagated through $\hat{f}$
$\mathbf{U_y} = \mathbf{C}\,\mathbf{U_x}\,\mathbf{C}^\top$
sensor and input uncertainties
$u(x_i)$, Type A or Type B
input correlations
covariances in $\mathbf{U_x}$; random and systematic components
propagation through the learned model
LPU or Monte Carlo (GUM-S1)
Measurement-model uncertainty u_{model}
$\hat{f}$ is an inexact estimate of f
$u_{\text{model}}^2 = u_{\text{training}}^2 + u_{\text{architecture}}^2 + u_{\text{calibration}}^2$
training-process variability
u_{training}: seed ensemble, Type A
model and architecture selection
$u_{\text{architecture}}$
approximation and generalization effects
$u_{\text{calibration}}$: reference values, validation
finite and non-representative training data
no term of its own: acts through the terms above